6. Covariance matrix of Two arrays.

program :-

```
import numpy as np.
data = np.array([[2,4,6],[1,3,5]]).

Cov-matrix = np.cov(data)

print("Data :\n", data)
print("Covariance matrix :\n", cov.matrix).
```

3. standard Deviation.

program:

```
import numpy as np
data = np.array ([10, 20, 30, 40, 60])
std_deviation = np.std (data)

print ("Data : ", data)
print ("standard Deviation : ", std_dev).
```

4. Variance.

program.

```
import numpy as np
data = np.array ([10, 20, 30, 40, 50])
variance = np.var (data)
print ("Data : ", data)
print ("Variance : ", Variance).
```

5. Covariance.

program:

```
import numpy as np
x = np.array ([2, 4, 6, 8])
y = np.array ([1, 3, 5, 7])

Covariance matrix = np.cov (x, y).
print ("Array x : ", x)
print ("Array y : ", y)
print ("Covariance : ", covariance - matrix [0][1])
```

Exp2 : Using python write numpy program & compile.

Date :

1. Expected value
2. Mean
3. Standard Deviation
4. Variance
5. Covariance
6. covariance matrix of two given arrays.

Aim :- To write and execute a python program using numpy to compute the expected value mean standard deviation, variance, covariance and covariance matrix for given datasets.

1. Expected Value :

program :

```
import numpy as np.
x = np.array ([1,2,3,4])
p = np.array ([0.1, 0.2, 0.3, 0.4])

expected_value = np.sum (x*p)
print ("Values : " x)
print ("Probabilities : ", p)
print ("Expected value : ", expected_value).
```

2. Mean :-

program :

```
import numpy as np
data = np.array ([10, 20, 30, 40, 50])
mean_value = np.mean (data)
print ("Data " : , data)
print ("Mean : ", meanvalue)
```